

Security Concepts

Module 1

Unit 2: The Big Issues



Learning Objective

LO1

Explain the following concepts, their examples in a given context:

- Cybercrime;
- Cyber harassment or cyber bullying;
- Cyber warfare;
- Cyber surveillance.

LO2

Discuss what makes cyber security difficult.



The Big Issues

Four Areas of Cyber Security

- Cybercrime;
- Cyber harassment or cyber bullying;
- Cyber warfare;
- Cyber surveillance.

Cybercrime



Cybercrime

Financial Theft

is the most widespread type of cybercrime

requires **little** or no **risk** to the thief :

- ✓ no guns
- ✓ masks
- ✓ getaway cars

and can deliver a significantly **greater reward**.

Downside: be an audit trail, indicating where the money came from and where it was transferred to.



Cybercrime

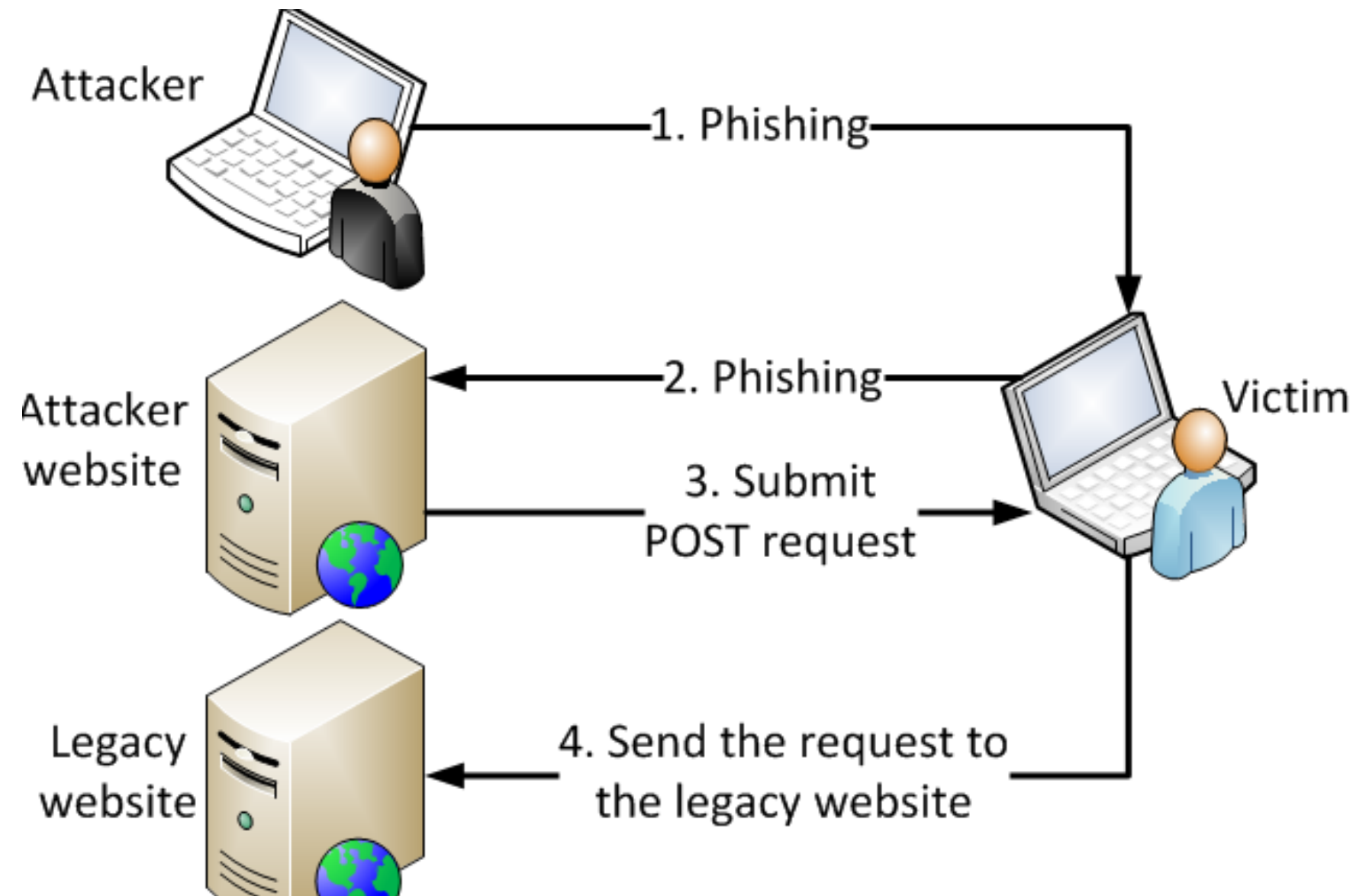
Website Defacement

some hackers will simply **deface** a website (usually its 'landing' page) in order to make their point

Often on:

- ✓ Political
- ✓ ethical
- ✓ environmental matters

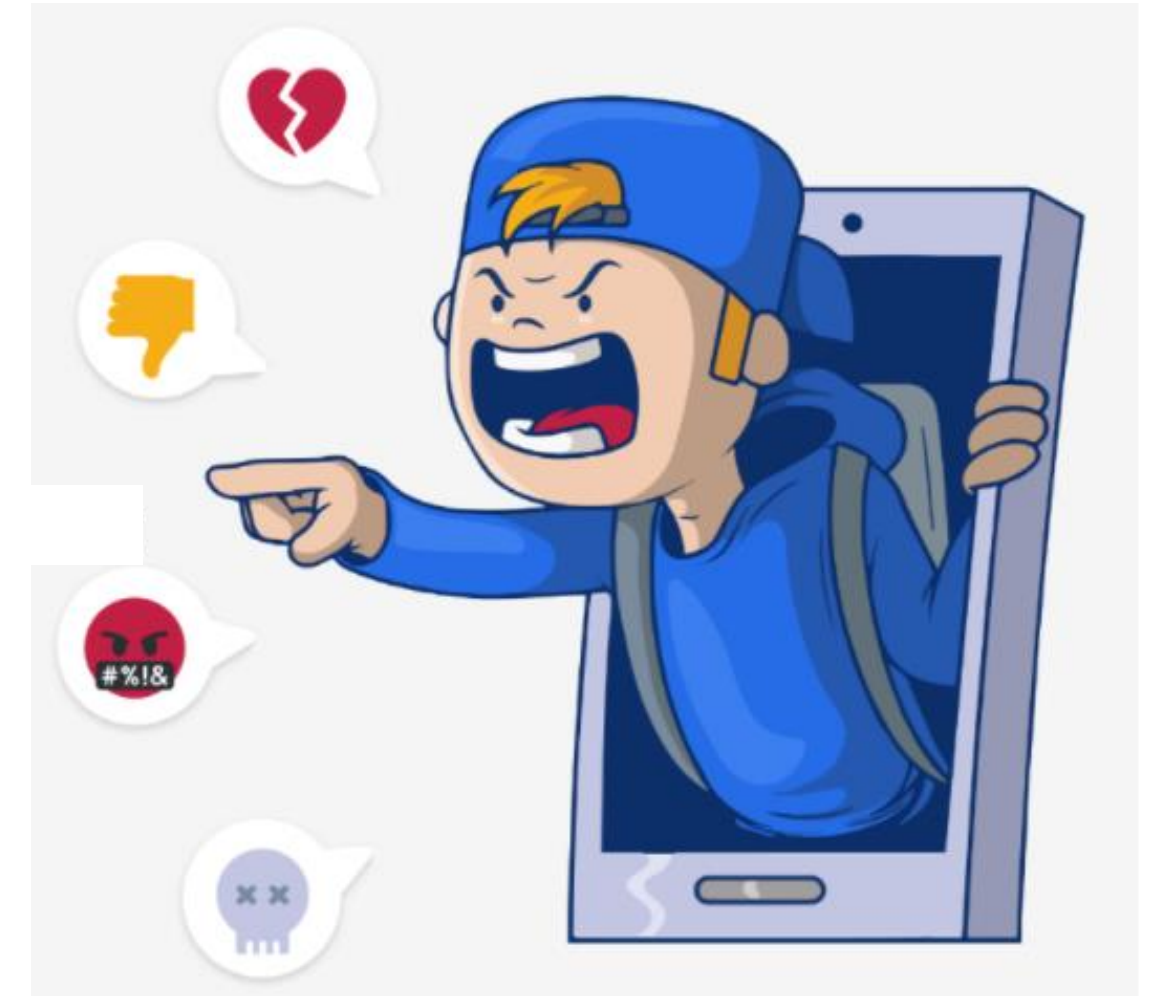
Your login credentials will be sent to the hacker instead of the site you normally access. **Be careful** and Always have means of **verifying** that the site you are browsing on is the **original site**.



CYBER HARASSMENT OR CYBER BULLYING

The act of harassing or bullying **a person or group of people** using cyber-based methods such as:

- ✓ social media
- ✓ text messaging
- ✓ and the like



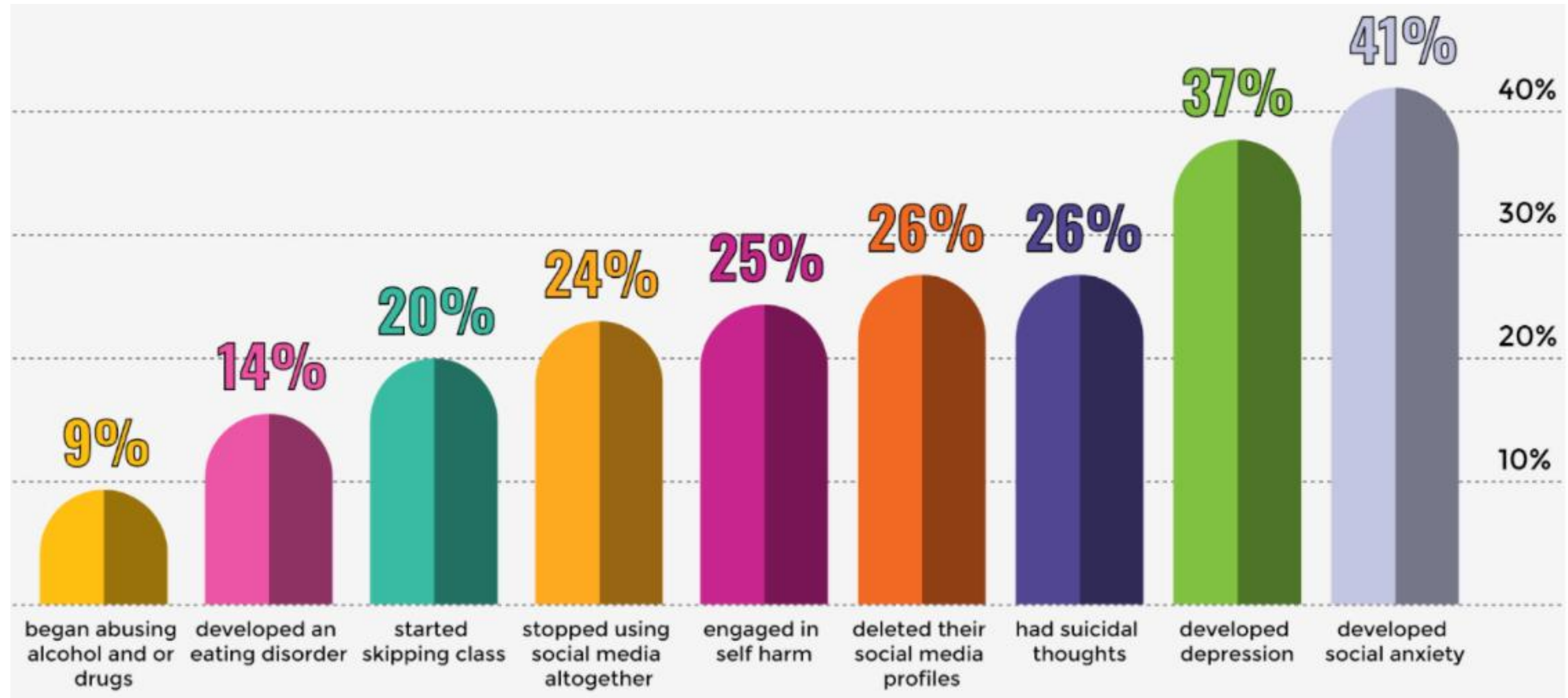
The difference between cyber harassment and cyber bullying

Cyber harassment: anyone or any organization can be the victim

Cyber bullying: generally refers to children as being the victims

CYBER HARASSMENT OR CYBER BULLYING

The Effects of Cyber Bullying



CYBER HARASSMENT OR CYBER BULLYING

Cyber Stalking

follow the **movements** and **activities** of their victim by **stealth**, and **not alert** them to the fact that someone is following them

they can still follow the **movements** and **activities** of their victim, but this time rather **more openly**, with the victim **being aware** they are being stalked, but usually without **knowing the identity of the stalker**.



CYBER HARASSMENT OR CYBER BULLYING

Cyber Trolling

Is a form of **verbal abuse** designed to **intimidate** or **offend** the victim

Confrontational or **abusive statements** online

Cyber trolls rarely make much effort to **hide** their identity



Cyber Warfare



- Process by which one **nation state** or **politically motivated group** conducts an attack against some aspect of another.
- A militant group could declare war on many nations.
- Cyber warfare adopts both **symmetric** and **asymmetric methods**

Cyber Warfare

Espionage

- Is the capability to **obtain secret information** without either the permission or the **knowledge of its owner**.
- Governments routinely **spy** on one another. They have done so for centuries.
- Espionage involves **agents** who place themselves in some **danger** by **operating in enemy territory**.



Cyber Warfare

Surveillance

- Focuses on keeping track of people's:
 - ✓ activities,
 - ✓ communications
 - ✓ and contacts
- It is more like an investigation into terrorism



Cyber Warfare

Sabotage

- is much **less risky** for its teams of saboteurs
- Stuxnet attack on the Iranian nuclear research programme.
- It was believed to have been conducted by a joint US–Israeli team
- developed software that would identify a very specific laboratory
- caused the **control system** to repeatedly **speed up then slow down the rotation** of the centrifuges



Cyber Surveillance



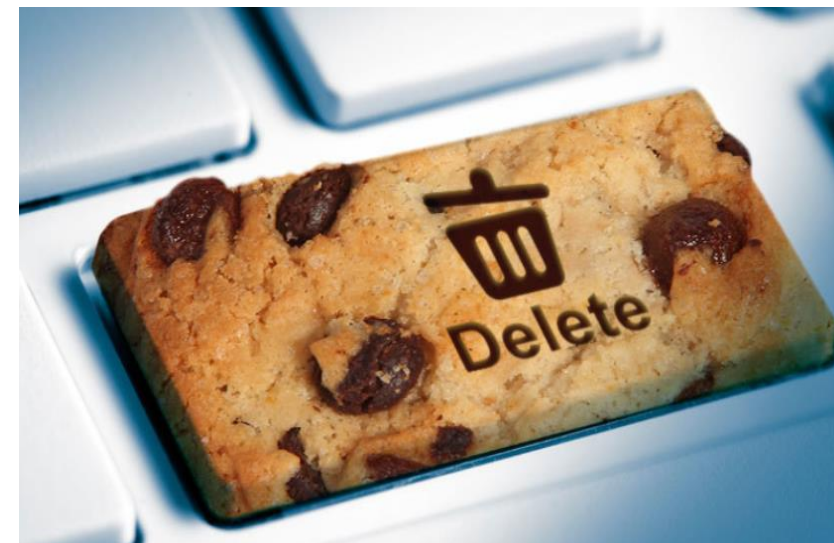
- Whether or not we are conscious of the fact, **we are continually under surveillance.**
- is usually associated with surveillance by the **security services.**
- is the **collection** and **use of data** about us by organizations with whom we interact on a daily basis.

Cyber Surveillance

Internet Search



Cookies



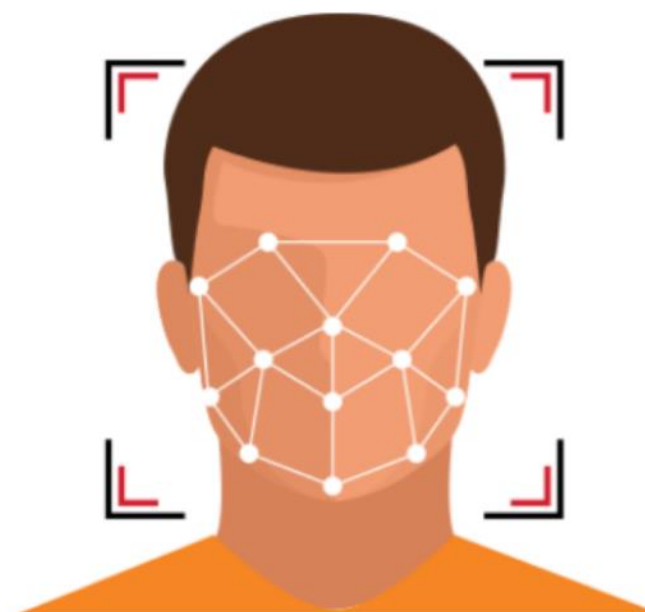
Email



Smartphones



Facial Recognition



Credit Cards



Why We Should Care

- It is nothing less than good practice to **manage risk**
- Customers have a right to expect organizations to **safeguard their information**
- organizations may need to be able to **demonstrate compliance** with national laws.
- Organizations should be able to **demonstrate good security practice** as a means of achieving competitive advantage

What makes Cyber Security Difficulty?

- Cyber security knowledge and skills
- Cyber security capabilities
- Cyber security standards and implementation

PREPARE FOR A
PRODUCTIVE YEAR

